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Gilford et al.

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(54) **STRAWBERRY PLANT NAMED ‘DRISCOLL SANIBEL’**

(50) Latin Name: *Fragaria×ananassa*
Varietal Denomination: **Driscoll Sanibel**

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See application file for complete search history.

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(57) **ABSTRACT**

This invention relates to a new and distinct variety of
strawberry named ‘Driscoll Sanibel.’ The variety is similar
to the varieties ‘Biscayne’ and ‘Key Largo.’ The variety is
distinguished from ‘Biscayne’ and ‘Key Largo,’ in
particular, by its smaller number of teeth on the terminal
leaflets, longer fruit length, greater number of achenes per
berry, greater average fruit yield, flat terminal leaflet margin
profile, and moderate resistance to *Verticillium*.

5 Drawing Sheets

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1. Latin name of the genus and species of the plant
claimed: The variety is botanically identified as *Fragaria×*
ananassa.

1.1 Variety denomination: The strawberry variety
denomination is ‘Driscoll Sanibel’.

2. BACKGROUND OF THE INVENTION

The new variety originated as a result of a controlled cross
between the strawberry plants ‘10D213’ (unpatented
variety) and ‘88E94’ (unpatented variety) in an ongoing
breeding program, and was discovered as a seedling in
Hillsborough, Fla. in 1999. The original seedling of the new
cultivar was asexually propagated by stolons in a Nursery in
Shasta County, Calif. Propagules were transplanted to a
controlled breeding plot in Hillsborough County, Fla., where
the variety was identified and selected for further evaluation.
‘Driscoll Sanibel’ was subsequently asexually propagated
and underwent further testing in Hillsborough County, Fla.
for three years. This propagation and testing has demon-
strated that the combination of traits disclosed herein which
characterize the new variety are fixed and retained true to
type through successive generations of asexual reproduc-
tion.

3. SUMMARY OF THE INVENTION

The present invention relates to a new and distinct variety
of strawberry named ‘Driscoll Sanibel.’ The variety is
botanically identified as *Fragaria×ananassa*. The new vari-
ety is distinguished from other varieties by a number of
characteristics as set forth in Tables 1 to 4.

4. COMPARISON TO SIMILAR VARIETIES

The varieties which we believe to be similar to ‘Driscoll
Sanibel’ from those known to us are ‘Biscayne’ (U.S. Plant
Pat. No. 12,186) and ‘Key Largo’ (U.S. Plant Pat. No.
8,649). There are several characteristics of the new variety
that are different from, or not possessed by ‘Biscayne’ and
‘Key Largo’. The new variety has a smaller number of teeth

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on the terminal leaflets, a longer fruit length, a greater
number of achenes per berry, a greater average fruit yield, a
flat terminal leaflet margin profile, and a moderate resistance
to *Verticillium*.

5 ‘Driscoll Sanibel’ is distinguished from its maternal par-
ent ‘10D213’ by its larger average berry size, less vigor, and
a more open plant canopy. ‘Driscoll Sanibel’ is distinguished
from its paternal parent ‘88E94’ by its larger average berry
size, improved shipping ability, and partial everbearing in
10 comparison to the full everbearing of ‘88E94’.

5. BRIEF DESCRIPTION OF THE DRAWINGS

15 The accompanying photographs show typical specimens
of the new variety, including fruit, foliage and flowers, in
color as nearly true as it is reasonably possible to make in
color illustrations of these characteristics.

FIG. 1 shows the upper side of the leaves of the plant.

FIG. 2 shows the whole the plant.

20 FIG. 3 shows the under side and underside of the flowers.

FIG. 4 shows a close-up of the fruit.

FIG. 5 shows the fruit in longitudinal cross-section.

6. DESCRIPTION OF THE NEW VARIETY

25 The following detailed description of the new variety is
based upon observations taken of plants and fruit grown in
Hillsborough County, Fla., U.S.A. This description is in
accordance with UPOV terminology. Observations of
30 ‘Driscoll Sanibel’, ‘Biscayne’ and ‘Key Largo’ were taken in
side by side comparison in the 2003–2004 winter season.
Plants for observation were harvested from McArthur,
Calif., and held in refrigerated storage until planting in
Hillsborough county, Fla. in October 2003. Plants were
35 grown in raised beds of soil under conditions typical of
commercial strawberry production in central Florida. Fruits
were harvested twice weekly for yield determination from
November 2003 to March 2004. Measurements of plant,
flower, and fruit characteristics were made in January 2004,

approximately four months after planting. Color designations, color descriptions, and other phenotypical descriptions may deviate from the stated values and descriptions depending upon variation in environmental, seasonal, climatic and cultural conditions. Colors are described and the most similar color designations are provided from The Royal Horticultural Society (R.H.S.) Colour Chart.

6.1 PROPAGATION

The new variety is principally propagated by way of stolons. Although propagation by stolons is presently preferred, other known methods of propagating strawberry plants may be employed.

6.2 CHARACTERISTICS OF THE NEW VARIETY

Information on the new variety is presented in Tables 1, 2, 3 and 4. In the tables, the flowers described are secondary flowers except where indicated. The fruit described is the secondary fruit on five month old plants. Fruit and flower measurements are an average of both primary and secondary fruit and flowers.

Table 1 provides information on the plant and fruit characteristics of the new variety 'Driscoll Sanibel' compared with characteristics of 'Biscayne' and 'Key Largo.' Table 2 provides additional information of the plant and fruit characteristics of the new variety 'Driscoll Sanibel' compared with characteristics of the varieties 'Biscayne' and 'Key Largo.' Table 3 provides reactions of the new variety to stresses, pests and diseases compared with reactions of the varieties 'Biscayne' and 'Key Largo.' Table 4 provides isozyme characteristics of the new variety as compared to that of the varieties 'Biscayne' and 'Key Largo.'

TABLE 1

QUANTITATIVE COMPARISON OF 'DRISCOLL SANIBEL', 'BISCAYNE,' AND 'KEY LARGO'			
	'Driscoll Sanibel'	'Biscayne'	'Key Largo'
<u>Plant Characteristics</u>			
Height of Plant (cm)	13.3	10.0	11.6
Spread of Plant (cm)	32.7	31.3	32.1
Number of Crowns	3.7	3.2	4.2
<u>Leaf Characteristics</u>			
Terminal Leaflet Length (cm)	7.7	7.6	8.4
Terminal Leaflet Width (cm)	8.1	8.1	8.0
Terminal Leaflet Length/Width Ratio	0.96	0.93	1.06
# Teeth/Terminal Leaflet	19.2	27	21.6
Color of upper side	131B medium green	137A light green	137A light green
Color of under side	142C light green	139C light gray green	139C light gray green
Petiole Length (cm)	11.9	9.2	11.5
Petiole Color	145D yellow green	142D yellow green	142D yellow green
Petiolule Length (mm)	6.5	5.5	6.3
Petiolule Diameter (mm)	2.1	2.2	1.9
Petiolule Color	145D yellow green	142D yellow green	142D yellow green

TABLE 1-continued

QUANTITATIVE COMPARISON OF 'DRISCOLL SANIBEL', 'BISCAYNE,' AND 'KEY LARGO'			
	'Driscoll Sanibel'	'Biscayne'	'Key Largo'
Bract Frequency	25% typically paired	83% typically paired	0%
Stipule Length (cm)	3.7	3.7	3.4
Stipule Width (cm)	1.8	2.0	1.5
<u>Stolon Characteristics</u>			
Anthocyanin color	53A red	59B red	
Diameter at bract (mm)	3.21	3.24	3.30
Avg. # of Daughter plants (2003 Nursery)	52	72	52
<u>Flower Characteristics</u>			
Petal Length (cm)	1.51	1.36	1.65
Petal Width (cm)	1.34	1.49	1.46
Petal Length/Width Ratio	1.13	0.91	1.13
Flower Diameter (cm)	3.05	2.76	2.72
Calyx Diameter (cm)	4.36	3.86	5.11
Sepal Length (mm)	17.0	15.3	21.5
Sepal Width (cm)	8.0	7.22	9.15
Petal Color	155C white	155C white	155C white
Receptical Color	149B yellow green	150A yellow green	150A yellow green
Anther Color	9A yellow	9A yellow	9A yellow
Fruiting Truss Length (cm)	12.9	8.7	13.6
<u>Fruit Characteristics</u>			
Fruit Length (cm)	5.8	4.8	4.8
Fruit Width (cm)	4.4	4.0	3.5
Fruit Length/Width Ratio	1.31	1.20	1.37
Average Berry Weight (g)	32.4	26.5	23.2
External Color	43A red	46B red	46B red
Internal Color	40C orange red	44A medium red	44B medium red
Achene Coloration	182A to 150C greyed red to yellow green	180B to 150C greyed red to yellow green	182B to 150A greyed red to yellow green
Achenes per berry	130	109	104
Achene weight (g)	0.00077	0.00046	0.00058
2003-2004 Fruit Yield (g/plant)	351	243	251

TABLE 2

QUALITATIVE COMPARISON OF 'DRISCOLL SANIBEL', 'BISCAYNE,' AND 'KEY LARGO'			
	'Sanibel'	'Biscayne'	'Key Largo'
<u>Plant</u>			
Habit	flat	flat globose	globose
Canopy Density	medium	medium	medium
Vigor	strong	strong	medium
<u>Leaf</u>			
Shape in cross section	slightly concave	concave	slightly concave
Interveinal blistering	medium to strong	weak	weak
Glossiness	medium	medium	medium
Number of leaflets	three only	three only	three only

TABLE 2-continued

QUALITATIVE COMPARISON OF ‘DRISCOLL SANIBEL’, ‘BISCAYNE,’ AND ‘KEY LARGO’			
	‘Sanibel’	‘Biscayne’	‘Key Largo’
Terminal leaflet margin profile	flat	revolute	revolute
Terminal leaflet shape of base	obtuse	rounded	slightly oblique
Terminal leaflet shape of teeth	rounded	rounded	rounded
Stipule pubescence	medium	medium	medium
Petiole pubescence	medium	medium	medium
Petiole pose of hairs	outward to downward	outwards	upward
<u>Stolon</u>			
Anthocyanin coloration	strong	strong	
Thickness	medium	medium	
Pubescence	medium	medium	
<u>Inflorescence</u>			
Position relative to foliage	beneath to level with	level	level to above
Diameter of calyx relative to corolla on secondary flowers	same size to larger	larger	much larger
Diameter of inner calyx relative to outer on secondary flowers	larger	same size	smaller
Spacing of petals	touching to overlapping	overlapping	overlapping
<u>Fruiting Truss</u>			
Attitude at first picking	prostrate	prostrate	semi-erect
<u>Fruit</u>			
Predominant shape	conical	conical	conical
Difference in shapes between primary and secondary fruits	slight	slight	slight
Band without achenes	narrow	narrow	narrow
Unevenness of surface	medium	weak	weak
Evenness of color	even	even	even
Glossiness	strong	strong	strong
Insertion of achenes	level with surface	level with surface	below surface
Insertion of calyx	level	level	set above
Pose of the calyx segments	spreading to reflexed	spreading	reflexed
Size of calyx in relation to fruit on secondary fruit	larger	larger	larger
Adherence of calyx	strong	strong	strong
Firmness of flesh	medium	firm	firm
Evenness of flesh color	slightly uneven	slightly uneven	slightly uneven
Distribution of flesh color	marginal and central	marginal and central	marginal and central
Hollow center size	medium	large	medium
Sweetness	strong	strong	medium
Texture when tasted	fine	fine	medium
Acidity	medium	medium	medium
Time of First Flowering in 2003–2004	Early November	Mid-November	Late November
Harvest Interval in 2003–2004	Late November to Mid-March	Early December to Mid-March	Late December to Mid-March
Type of Bearing	partially everbearing	partially everbearing	partially everbearing

6.3 REACTION TO STRESS, PESTS, AND DISEASE

TABLE 3

REACTIONS TO STRESS PESTS AND DISEASES FOR ‘DRISCOLL SANIBEL’, ‘BISCAYNE,’ AND ‘KEY LARGO’			
	‘Driscoll Sanibel’	‘Biscayne’	‘Key Largo’
<u>Reaction to Pests</u>			
<i>Tetranychus urticae</i>	susceptible	susceptible	moderately susceptible
<i>Lygus hesperus</i>	susceptible	susceptible	susceptible
<u>Reaction To Diseases</u>			
Botrytis fruit rot	susceptible	susceptible	susceptible
Powdery mildew	susceptible	moderately susceptible	susceptible
<i>Verticillium</i> wilt	susceptible	susceptible	susceptible
Strawberry Mottle Virus	moderately resistant	moderately resistant	moderately resistant
<i>Xanthomonas fragariae</i>	moderately susceptible	moderately susceptible	moderately susceptible

6.4 ISOZYME ANALYSIS

In addition to the morphological description above, the new cultivar ‘Driscoll Sanibel’ has been analyzed to obtain an indication of its genetic makeup to provide further means for identifying the new variety and distinguishing it from other somewhat similar and/or related strawberry varieties. Specifically, leaf samples of ‘Driscoll Sanibel’, ‘Biscayne’ and ‘Key Largo’ were analyzed by electrophoresis for isozyme patterns of the enzymes phosphoglucosomerase (“PGI”), leucine aminopeptidase (“LAP”) and phosphoglucosomutase (“PGM”). See J. Amer. Soc. Hort. Sci. 106:684–687. Isozyme characterization of the three varieties is presented in Table 4, with the letters representing the banding patterns for each enzyme as designated in the above-identified article.

TABLE 4

ISOZYME ANALYSIS FOR ‘DRISCOLL SANIBEL’, ‘BISCAYNE,’ AND ‘KEY LARGO’			
Locus	‘Driscoll Sanibel’	‘Biscayne’	‘Key Largo’
PGI	A1	A1	A1
LAP	B3	B3	B3
PGM	C2	C4	C4

What is claimed:

1. A new and distinct variety of strawberry plant, substantially as shown and described.

* * * * *



FIG.1



FIG.2

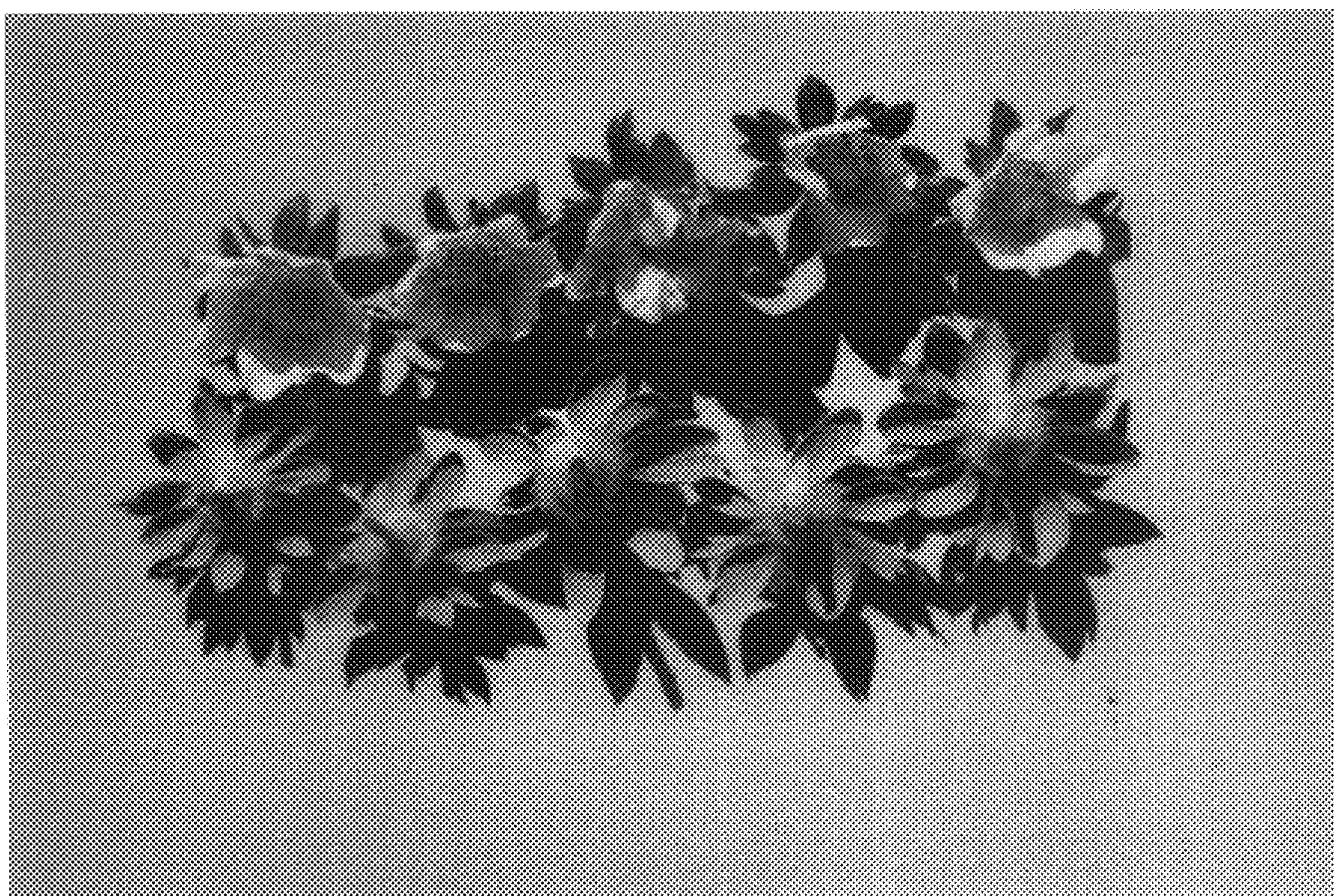


FIG.3



FIG.4



FIG.5